

AR-based Dynamic Sandbox for Hydraulic Earth-rock Dam Break

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ABSTRACT

VR/AR-based hydraulic scenes holds significant value for geological disaster display. However, the river dynamics and dam break models prevalent in computational fluid dynamics (CFD) and hydraulics rely largely on pure numerical simulations, lacking intuitive visual representation. In contrast, current AR-based physical simulation methods often fail to incorporate hydraulic parameters adequately, falling short of the accuracy needed for real hydraulic models. To bridge this gap, this paper proposes integrating a precise numerical river dynamics model into a visual fluid-solid interaction simulation framework, which achieves a balance between a realistic visual simulation of dam breaks and the numerical accuracy aligned with river dynamics models, while maintaining high computational efficiency.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction devices—; Computing methodologies—Computer graphics—Animation—Physical simulation

1 INTRODUCTION

There is a significant divide between computational fluid dynamics simulations of river dam breaks and virtual reality visualizations of fluid-structure interactions. The former focuses on precise numerical computations of real river dynamics and hydraulics, whereas the latter prioritizes the visual impact and the efficiency of user interactions within the virtual environment. Employing interactive AR sand tables to simulate and dynamically represent flood evolution and earth-rock dam failures is extremely important for disaster preparedness and emergency response training. The primary challenge lies in reconciling the accuracy of hydrodynamic models with the speed and interactivity of graphical fluid models (such as FLIP, MPM, SPH, etc.).

To bridge the gap between numerical computation and virtually visual simulation, we propose a novel framework that integrates river dynamics and computational fluid dynamics (CFD) models within the realm of virtual reality simulations. This integration is designed to equip visual fluid simulations with the ability to accurately reflect critical aspects of soil mechanics and hydraulic properties, such as the forces exerted on dam structures and the fracture infill process. As a result, this approach aims to preserve the numerical integrity of hydraulics and river dynamics while facilitating efficient and real-time simulations of the behavior of dams of the river and the earth. To realize this ambitious goal, we opt to employ the Material Point Method (MPM) as the cornerstone of our simulation framework, enabling us to model both the fluid dynamics and the solid mechanics of the earth-rock dam in a unified manner. To simulate interactive dam breaches, we incorporate a criterion based on river dynamics for the onset of soil mass erosion to simulate the onset of collapse and the subsequent breaking of the dam. This method not only

enhances the realism of virtual sand table simulations but also paves the way for innovative VR-based modeling techniques in large-scale hydraulic engineering endeavors.

2 VISUAL MODEL FOR HYDRAULIC EARTH-ROCK DAM BREAK

We use the MPM model to simulate the interaction between the liquid and the earth-rock dam. MPM excels at simulating the intricate coupling between material fluids and structures, offering a more nuanced depiction of interactions among liquids, gases, and solids. It can simulate the movement of materials, stress deformation, and other dynamic phenomena in computer graphical applications. However, the simplified physical solution process still ignores many important properties and boundary conditions required by soil mechanics and hydraulics compared to the CFD model. To this end, to enable MPM to process with the accuracy of numerical calculations, we introduce an additional critical start condition for earth-rock particles.

2.1 Scrolling Start Velocity for Earth-rock Dam Break

The earth-rock dam is initially modeled as a static solid. For each earth-rock particle, whether it is washed out or not depends on the criticality condition we introduce. As shown in Fig. 1, the additional force of the earth-rock particle includes three components: lift F_L , drag F_D and gravity W . The critical condition for rolling start (the particle is released by the constraint of the rigid body) is the following.

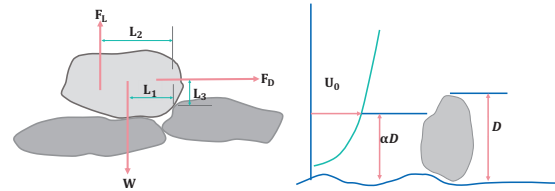


Figure 1: Stress on earth-rock particle. Figure 2: Flow velocity distribution near the particle.

$$W \times L_1 = F_L \times L_2 + F_D \times L_3, \quad (1)$$

where $W = \frac{1}{6}(\gamma_s - \gamma)\pi D^3$, $F_L = C_L \frac{\pi D^2}{4} \frac{\rho U_0^2}{2}$, and $F_D = C_D \frac{\pi D^2}{4} \frac{\rho U_0^2}{2}$. We set L_1 , L_2 , and L_3 are equal to the radius of MPM particle for simplification. We adopt an exponential formulation with the time-averaged flow velocity distributed along a vertical line:

$$u(y) = (1+m)\bar{U}\left(\frac{y}{h}\right)^m, \quad (2)$$

where as shown in Fig. 2, the velocity U_0 is defined as:

$$U_0 = (1+m)\alpha^m \bar{U}\left(\frac{D}{h}\right)^m. \quad (3)$$

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Table 1: The critical starting mean flow velocity U_c (m/s) obtained by the Shamov formula.

radius(mm)	h=0.2m	h=1m	h=10m	h=30m
0.05	0.13	0.17	0.25	0.30
0.5	0.28	0.36	0.53	0.64
5.0	0.60	0.78	1.15	1.38
50.0	1.29	1.69	2.48	2.98

Thus, we get the scrolling start velocity:

$$U_c = m \sqrt{\frac{\gamma_s - \gamma}{\gamma}} g D \left(\frac{h}{D}\right)^{1/6}, \quad (4)$$

where m is a constant. This equation is called the "exponential starting velocity formula". The starting velocity calculated by this kind of formula is directly related to the depth of the water. Among them, the Shamov formula is the most widely used, where m is set to 1.14. Average rolling start velocity values under different particle sizes and different water depths are shown in Table 2.1.

2.2 Evolution of Earth-rock Dam Outburst

Drawing upon the concept of rigid-fluid interaction as described by Carlson et al. [1] and Gao et al. [3], we employ the Material Point Method (MPM) to uniformly represent fluids and solids of earth-rock dams. Initially, we assign distinct labels to differentiate between the two types of materials.

For the particles representing the earth-rock dam, we treat them as static solids that interact with the fluid. Once the velocity of the fluid exceeds the critical threshold U_0 , a parameter previously discussed, we remove the solid constraint on the individual particles of the dam. This signifies that the particle has been eroded and carried away by high-speed fluid flow.

3 REAL-TIME RENDERING IN AR PHENOMENON

To facilitate real-time reconstruction of visual AR-based scenes, we apply distinct surface reconstruction and rendering techniques tailored to different material properties. Considering that rigid objects, such as a static sandbox and the initial earth-rock dam, do not experience deformation, and the surfaces of hand models are automatically tracked by the device, surface reconstruction is rendered unnecessary for these elements. Therefore, we directly use the original mesh models for rendering, which includes the integration of material textures and shaders.

For fluids, which undergo significant deformations and exhibit unstable surfaces, we utilize screen-space surface reconstruction techniques. Specifically, we implement anisotropic screen-space surface reconstruction for real-time rendering, as detailed in the work by Deng et al. [2].

Regarding solid particles that are released from the dam model, we streamline the process by reclassifying the particles' labels from solid to fluid prior to each solid computation step. This reclassification allows the particles to bypass solid reconstruction and directly engage in screen-space computations and rendering activities.

4 EXPERIMENTAL RESULTS

To evaluate the effectiveness of our method, we applied it to simulate the dam failure event at Izmir Urkme in Turkey, which has experimental data available for a distorted physical model provided by Guney et al. [4]. The scaled model used in these experiments has a horizontal ratio of 1:150 and a vertical ratio of 1:30.

Using the dam break module of the HEC-RAS hydrodynamic model, we conducted simulations. As depicted in Fig. 3, the flow curve generated by our method is juxtaposed with both the measured flow curve and the numerical simulation curve from the distorted

physical model. The comparative results are presented in the figure below.

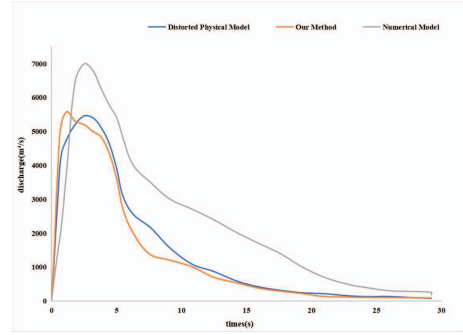


Figure 3: The visually dam breaking phenomenon illustrated in AR HMD.



Figure 4: The visually dam breaking phenomenon illustrated in a HoloLens HMD.

The visual experimental results are presented in Fig. 4, which depicts the dam breach scene as realized using a HoloLens HMD. The numerical method employs a statistical model that draws on historical dam failure data, but does not account for variables such as specific dam types, materials, or the physical mechanisms of dam failure. Consequently, there are noticeable disparities in the results. However, the overarching trends in dam breakage and the corresponding curves show similarity, suggesting that our AR-based method provides a useful reference for such scenarios.

5 CONCLUSION

We have successfully integrated a numerical model for simulating earth-rock dam failure, rooted in river dynamics, into the VR/AR domains. This integration enables the vivid visualization of computational results, offering an immersive experience that closely mirrors real-world scenarios of dam failure. Although the experimental results and numerical accuracy have room for further optimization, our initial experimental results demonstrate the potential of our approach for visualization and VR/AR display of hydraulic simulation. It provides a new idea for visual modeling and simulation for numerical water resource and geological computations.

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